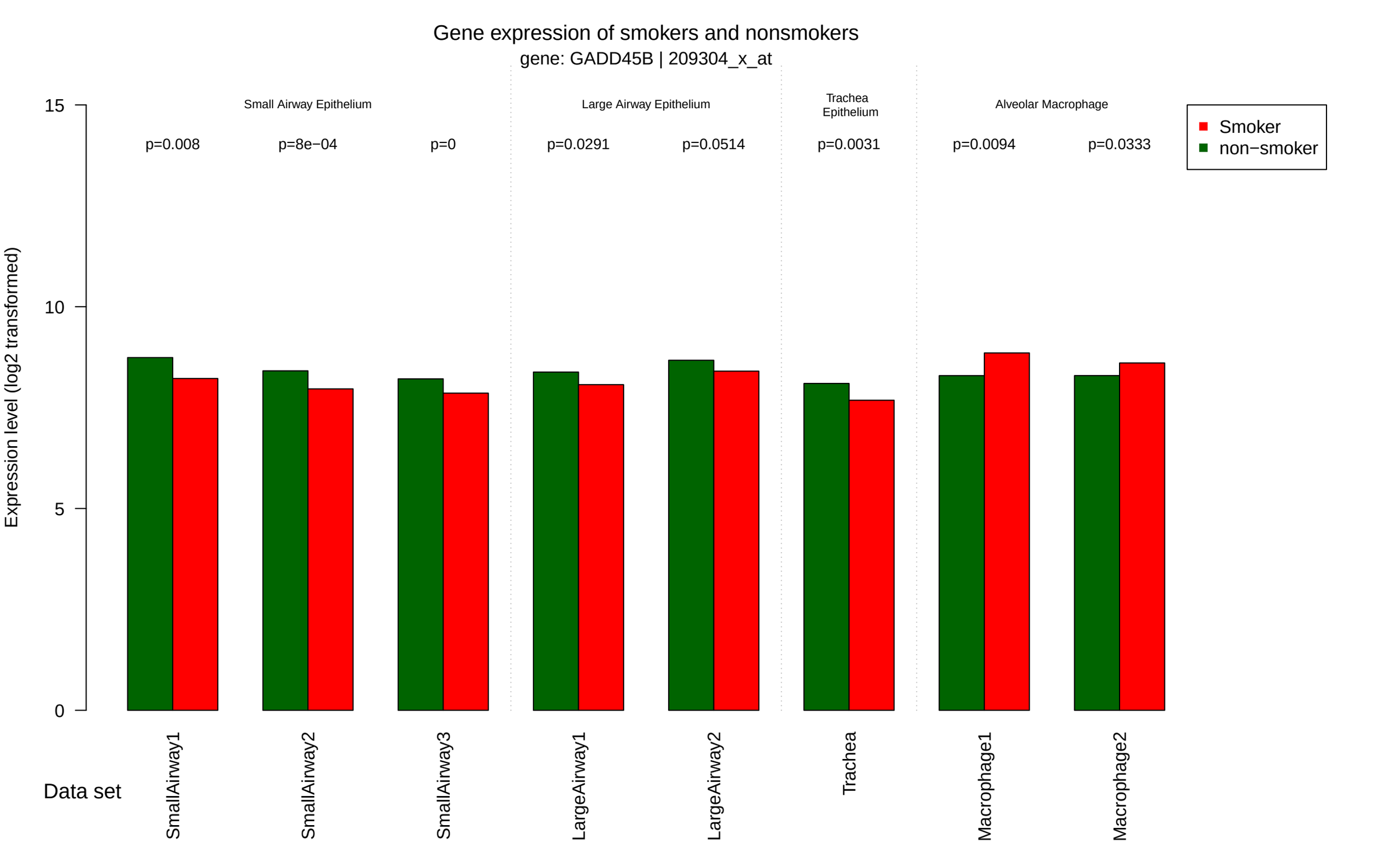
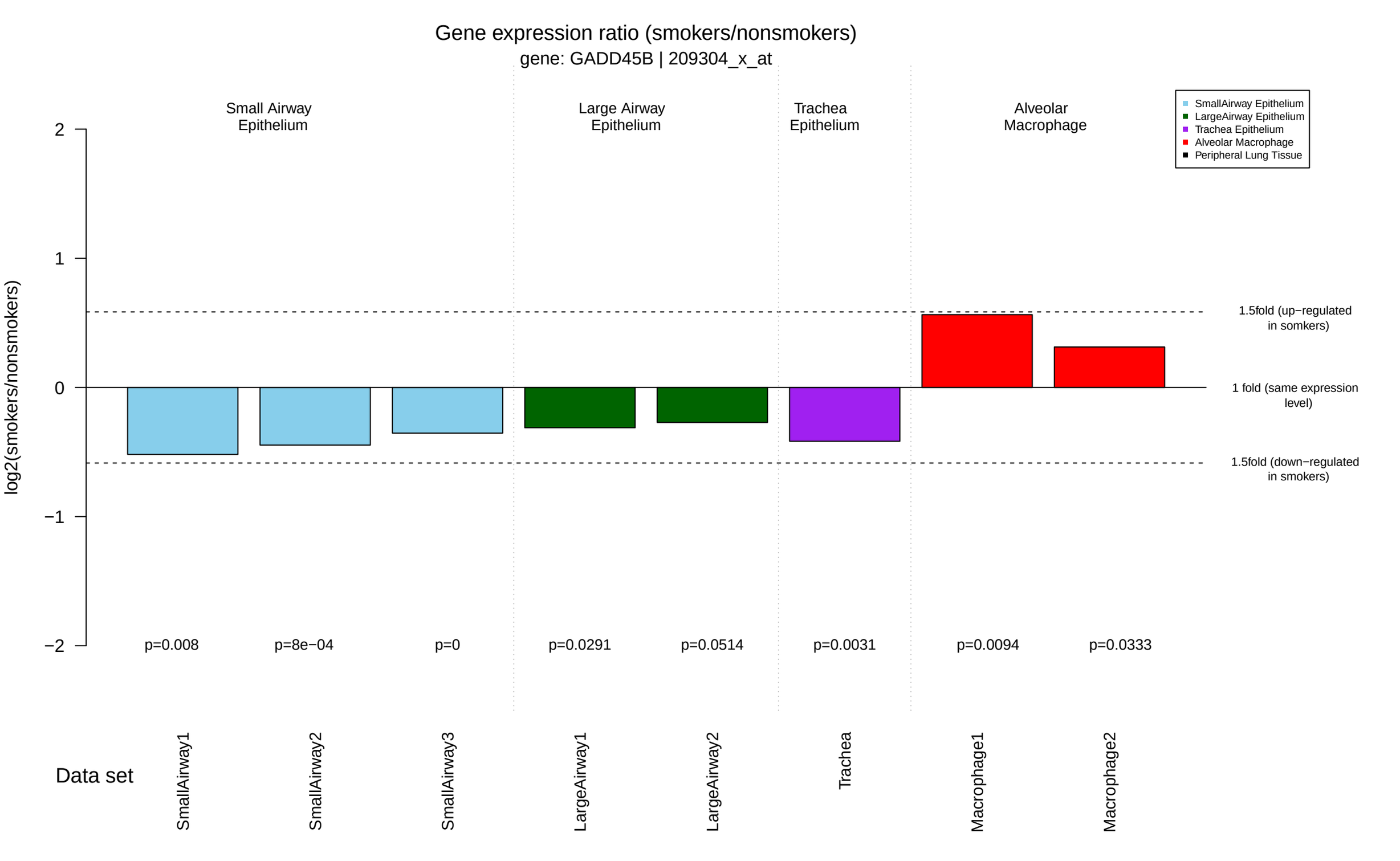
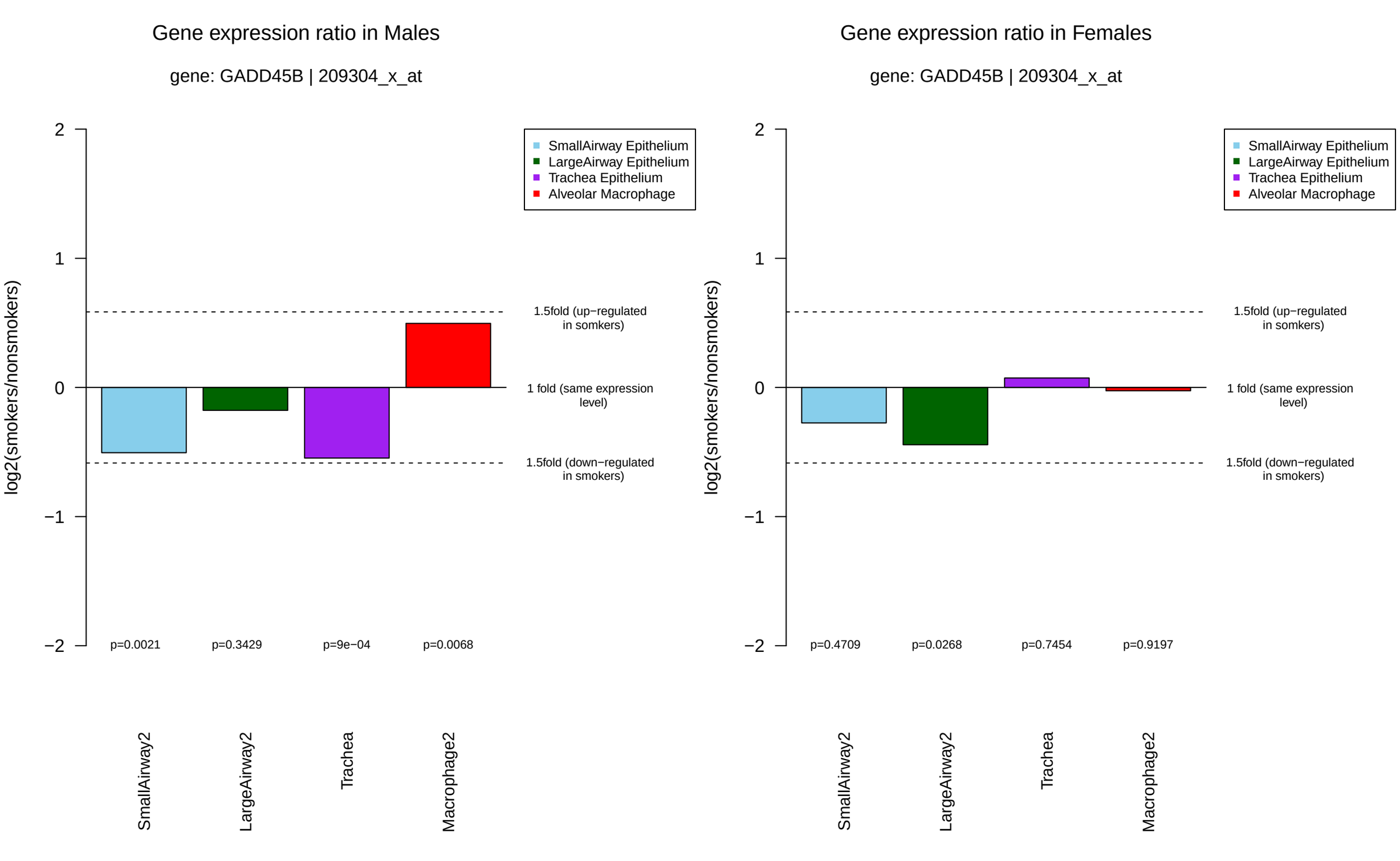
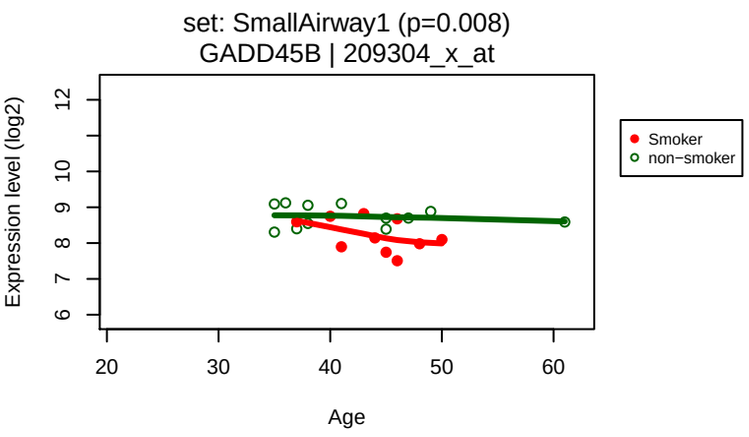
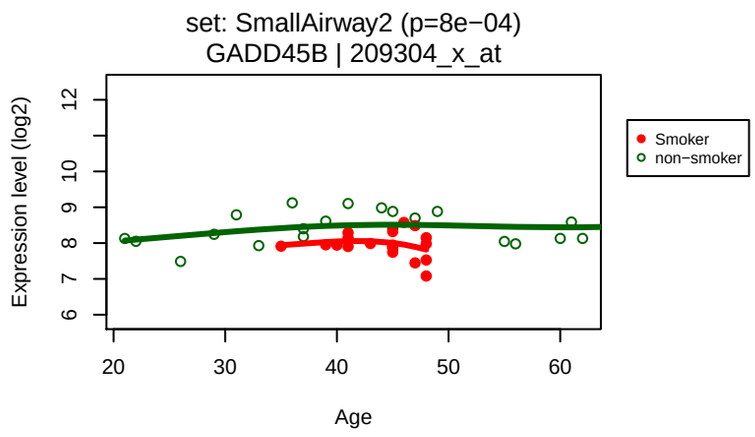
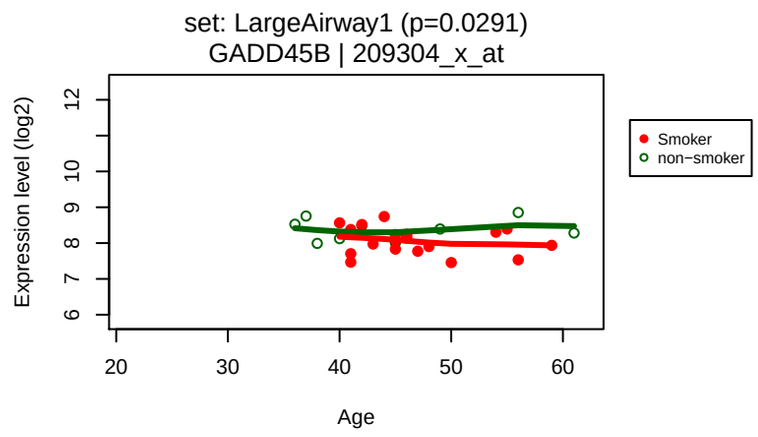

Gene expression of smokers and nonsmokers
gene: GADD45B | 209304_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.008
p=8e−04
p=0
p=0.0291
p=0.0514
p=0.0031
p=0.0094
p=0.0333
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

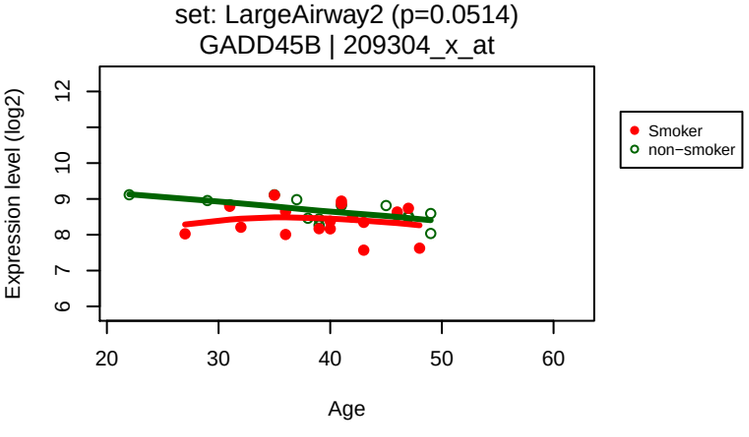

Gene expression ratio (smokers/nonsmokers)
gene: GADD45B | 209304_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.008
p=8e−04
p=0
p=0.0291
p=0.0514
p=0.0031
p=0.0094
p=0.0333
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

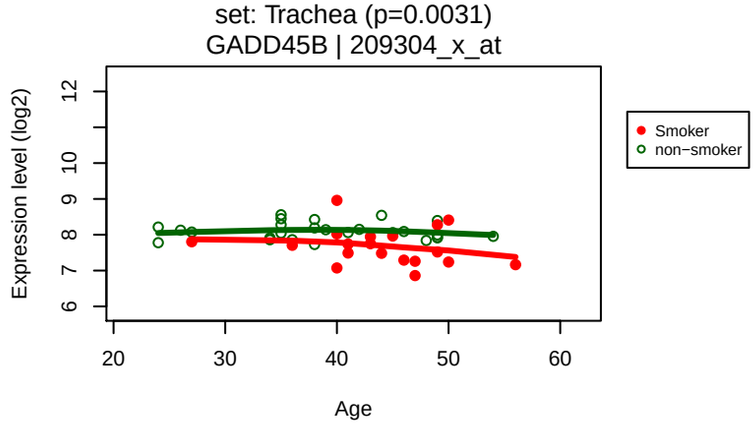

Gene expression ratio in Males
gene: GADD45B | 209304_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0021
p=0.3429
p=9e−04
p=0.0068
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GADD45B | 209304_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4709
p=0.0268
p=0.7454
p=0.9197
SmallAirway2
LargeAirway2
Trachea
Macrophage2

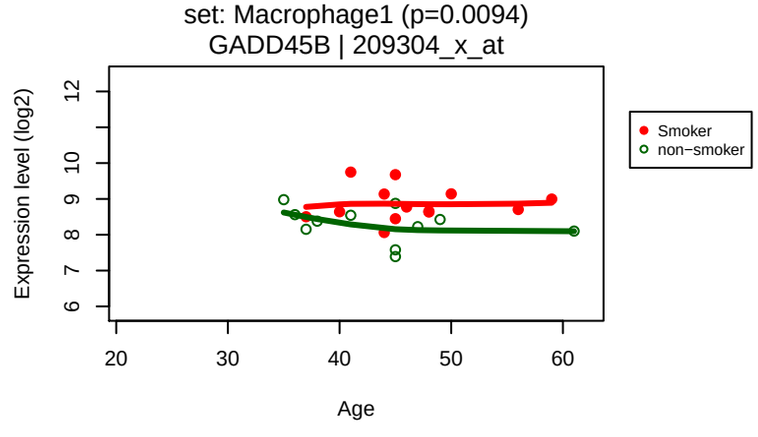

set: SmallAirway1 (p=0.008)
GADD45B | 209304_x_at
Expression level (log2)
Age
Smoker
non−smoker

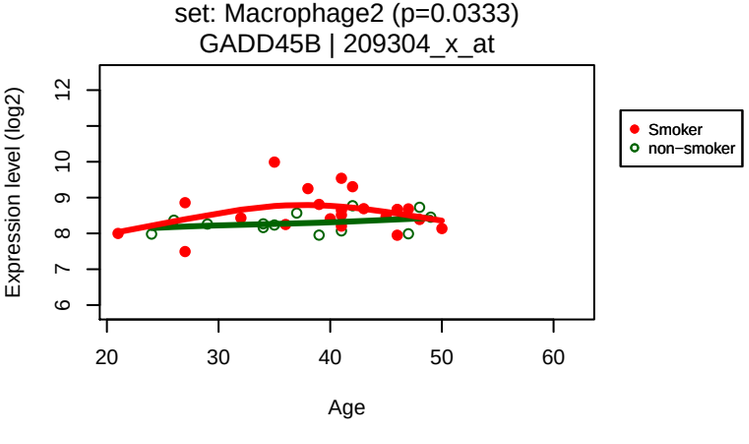

set: SmallAirway2 (p=8e−04)
GADD45B | 209304_x_at
Expression level (log2)
Age
Smoker
non−smoker

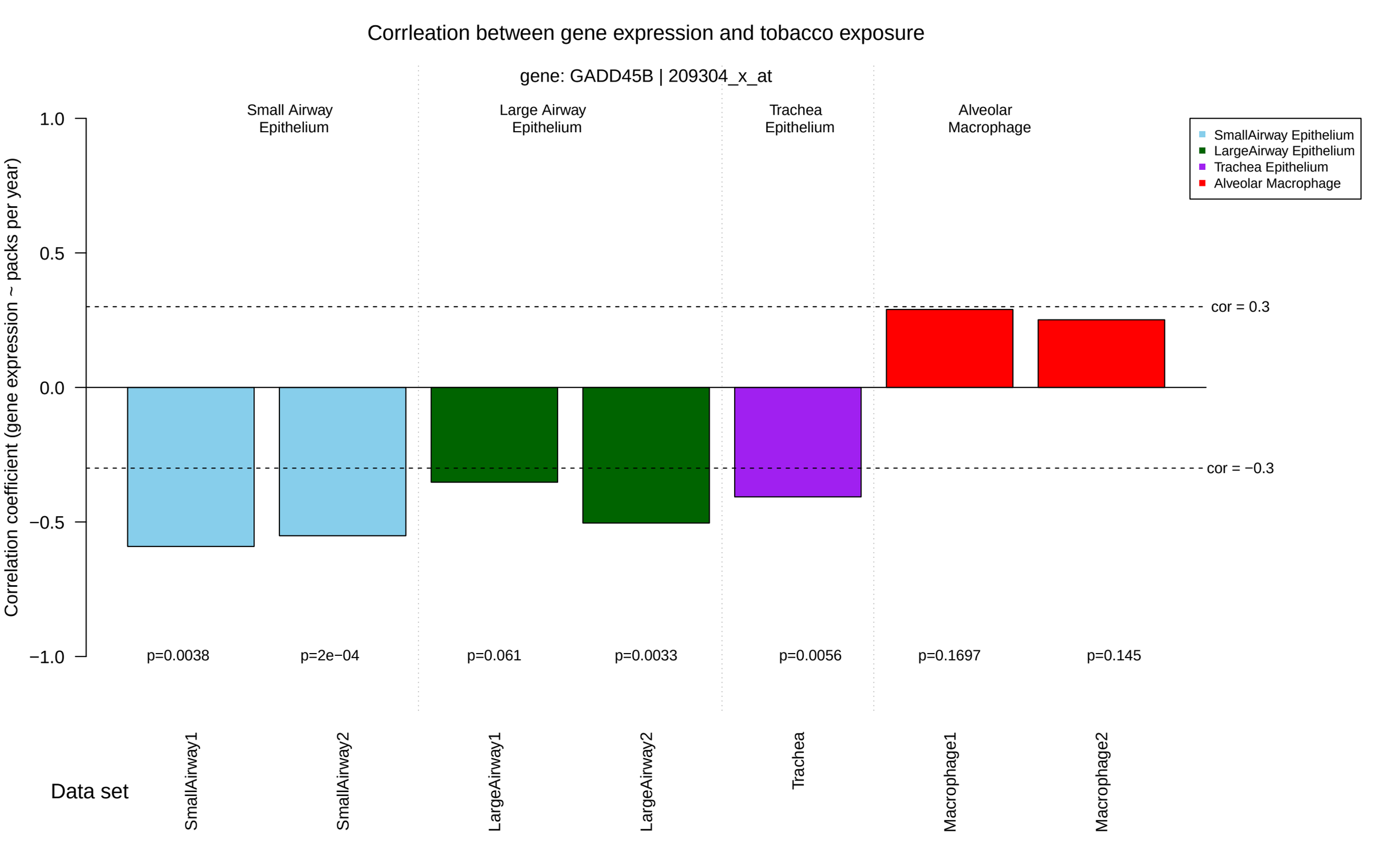

set: LargeAirway1 (p=0.0291)
GADD45B | 209304_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0514)
GADD45B | 209304_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0031)
GADD45B | 209304_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0094)
GADD45B | 209304_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0333)
GADD45B | 209304_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GADD45B | 209304_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0038
p=2e−04
p=0.061
p=0.0033
p=0.0056
p=0.1697
p=0.145
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2